

Coherent Compounding for Photoacoustic Tomography

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Abstract— Circular-view detector configurations provide higher image quality in a photoacoustic (PA) imaging system than other configurations. The image quality degrades, and streak artifacts appear in the reconstructed image as the number of detectors decreases. Here, we present a coherent compounding strategy for suppressing streak artifacts. Parts of the reconstructed image that refer to the excited PA absorbers and the artifact are separable by reconstructing several sub-images and combining the reconstructed images. The numerical results showed that the presented method suppresses artifacts and improves the quality of reconstructed images than the conventional method. The peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) of the presented method reaches 43 dB and 0.73, respectively, which results, in a 13 dB and 2.5-fold improvement of the PSNR and SSIM, respectively, compared to the conventional method.

Keywords—Photoacoustic, Circular-view, Image reconstruction, Compounding, Tomography.

I. INTRODUCTION

A laser source at first illuminates the biological tissue in a photoacoustic (PA) imaging (PAI) system. Then, the biological tissue absorbs light energy based on its optical absorption coefficient, and thermal expansion occurs which results in generating a PA wave. After that, the generated PA wave propagates in all directions (solid angles), and acoustic transducers (ATs) record the PA wave [1, 2]. The more solid angles of the PA wave recorded by the ATs, the greater the quality of the reconstructed image [3]. According to previous researches, there are two basic AT configurations for most PAI systems: (1) planar and (2) circular configuration [4, 5]. As a result, among several AT configurations for a PAI system, circular view, particularly full circular view (FCV), provides higher quality reconstructed images than planar and circular limited-view [6, 7].

Although the FCV configuration provides a higher-quality reconstructed image than other configurations, using a large aperture AT or an insufficient number of acoustic detectors degrades the image quality [8, 9]. Moreover, ATs have a limited frequency bandwidth. As a result, the image quality drops as the ATs record fewer frequency components of the PA wave than in the infinite bandwidth scenario [9]. In recent years, many researches have been reported to expand the bandwidth of AT synthetically and to improve the resolution and contrast in systems utilizing large aperture ATs [10-12]. Here, we concentrate on FCV scenarios utilizing a few acoustic detectors.

When low-density detection points (AT positions) are employed in the FCV-PAI system, aliasing and streak artifacts appear in the reconstructed image [13]. The minimal number of ATs necessary to avoid aliasing and streak artifacts may be estimated using the radius of the area of interest (ROI) and the upper frequency of ATs [14]. Several studies have been published to diminish aliasing and artifacts in low-density FCV-PAI systems. To reduce streak artifacts in PAT, Hu, et al. [15] developed a spatiotemporal antialiasing technique. They used a radius-dependent low-pass filter and spatial interpolation. While this approach reduces streak artifacts, it results in blurring. To increase the quality of images, Cai, et al. [16] suggested an adaptive back-projection technique. The characteristics of the tissue determine the parameters of their reconstruction technique. As a result, obtaining a high-resolution image may be difficult in some instances. To lessen artifacts in FVC-PAI systems, coherence-based weightings are proposed [17-19]. Moreover, to increase the quality of images and minimize artifacts and background noise, some model-based reconstruction techniques are presented [20-22]. Several iterative reconstruction techniques are also proposed for enhancing the quality of reconstructed images. Iterative algorithms, on the other hand, are often time-consuming. Deep learning (DL) approaches for improving contrast and image quality are also extensively addressed [23]. Awasthi, et al. [24] presented a deep neural network that helps to minimize noise and increase the signal-to-noise ratio (SNR). Lu, et al. [25] reported an unsupervised artifact removal methodology. [26, 27] suggested a deep learning approach to remove streak artifacts in sparse scenarios. Although DL algorithms increase the quality of images, their efficacy in *in-vivo* applications is limited by a lack of substantial experimental datasets.

In this paper, we describe a **compounded weighting function** technique for improving image quality under conditions containing low-density detection points. We split ATs into multiple groups to reconstruct several sub-images. Then, for each sub-image, we reconstruct it by selecting a group of ATs. The coherent compounding of these sub-images improves the image quality and eliminates streak artifacts. Section II describes the proposed strategy in further detail. Section III contains the results and discussion, while Section IV provides the conclusion.

II. MATERIALS AND METHODS

In order to achieve acceptable image quality in a circular-view PAI system, the distance between detection locations must be more than half of the acoustic detector's center frequency. As a result, the required minimum number of detection points (N) is as follows [14]:

$$N = \frac{4\pi Rf}{c}, \quad (1)$$

where, R is the scanning radius of the system, c is the mean of the sound speed in the media, and f is the central frequency of the detector. Therefore, for a typical breast/brain PAI system with $R = 10$ cm, $f = 2$ MHz and $c = 1500$ m/s, N would be ~ 1650 .

As previously stated, when PA absorbers with high initial pressure are excited in the ROI, they generate PA waves in all directions. Then, all ATs record their PA wave. The recorded PA wave will be back-projected as a circular arc in the ROI by running the back-projection reconstruction method for each detector position. As a result, back-projecting the recorded PA signal of each AT arc-shaped artifact, which correspond to high-intensity PA absorbers, would appear. As we employ independent ATs to reconstruct each sub-image, we expect the streak artifacts to appear in various places for different sub-images. As a result of the coherent compounding of these sub-images, we expect the primary components referring to PA absorbers to be strengthened and streak artifacts to be suppressed.

The detailed workflow technique of the proposed method is represented in Fig. 1 and described in the following sections.

Step 1: Reconstructing an image utilizing all of the ATs' data. At first, we perform a conventional delay-and-sum algorithm to all detectors' data to reconstruct the initial image [28].

Step 2: Defining the number of sub-images and choosing the positions of ATs to reconstruct each sub-image. In order to reconstruct sub-images, we define the number of sub-images (NSI). The NSI is an integer value that represents the overall number of sub-images. Also, the NSI is the sparsity number (SN) of ATs to reconstruct each sub-image. The required number of ATs to reconstruct each sub-image would be obtained by dividing the total number of ATs by SN. The position of ATs used for reconstructing each sub-image is independent of the positions of other sub-images. As a result, we select ATs with step size SN for each sub-image. For example, if we define $NSI = SN = 3$, the corresponding ATs for reconstructing the first sub-image are 1, 4, 7, 10,...; the corresponding ATs to reconstruct the second sub-image are 2, 5, 8, 11,...; and the corresponding ATs to reconstruct the third sub-image are 3, 6, 9, 12,....

Step 3: Performing delay-and-sum algorithm to reconstruct sub-images. After choosing the corresponding ATs for each sub-image, we reconstruct all sub-images by back-projecting the ATs' data into arc-shaped signals in the entire ROI. At the end of this procedure, several low-quality sub-images are reconstructed.

Step 4: Creating a signed image by calculating the sign value for each pixel in each sub-image. Compounding these sign images at each pixel is a constructive sum in structural

information locations and a destructive sum in artifact locations because the reconstructed sub-images are identical in parts that refer to structural information (position of the PA absorbers) and different in parts that refer to streak artifacts and aliasing.

Step 5: The final image is obtained by multiplying the created compounded sign image by the initial reconstructed image. In the final image, we expect that the streak artifacts are minimized and the PA absorber spots are strengthened.

The following section presents numerical analysis to evaluate the proposed methodology.

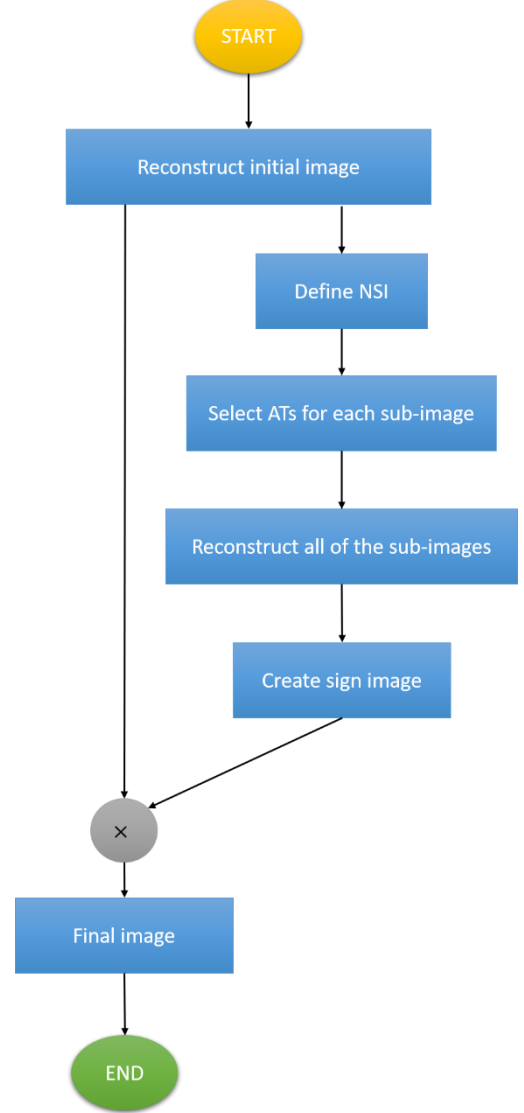


Fig. 1. The proposed method's workflow diagram.

III. NUMERICAL STUDY AND RESULTS

A. Numerical Study

The raw PA signals were synthesized using the k-Wave MATLAB package [29]. The Derenzo phantom and the arteries phantoms were synthesized to assay the proposed method. Also, 360 acoustic detectors were placed around the ROI with a radius of curvature of 52 mm in the media to record the PA signals at a sampling frequency of 25 MHz. After storing the PA signals, all of them were passed through a Gaussian band-pass filter with center frequency and

fractional bandwidth of 6.25 MHz and 70%, respectively. Moreover, the grid size of the media was considered 0.01 mm in both X-Y directions and the empty media of the phantom was synthetically filled with water with a speed of sound of 1500 m/s. For quantitative evaluation, we measured the structural similarity index (SSIM) and peak signal-to-noise ratio (PSNR) in the reconstructed images obtained by the conventional and proposed methods.

B. Numerical Results

1) Case 1: Derenzo phantom

Figure. 2(a) shows the Derenzo phantom. The yellow dashed circles in Fig. 2 indicate the position of ATs. The reconstructed image with the conventional method (one sub-image) and the proposed method using 35 sub-images are shown in Fig. 2(b, c), respectively. Several unwanted artifacts appeared in the reconstructed image obtained with the conventional approach, especially near the PA absorbers [compare white arrows in Fig. 2(b)]. In the proposed method, NSI/SN was considered 36, and ten ATs' PA signals were used to reconstruct each sub-image. The proposed method suppresses unwanted artifacts in crowded regions and improves the quality of the reconstructed image compared to the conventional method [compare arrows in Fig. 2(b, c)]. TABLE I reports the quantitative values of the conventional and proposed method. The conventional method's SSIM and PSNR values were measured at 0.29 and 30 dB, respectively. The proposed method increases the SSIM by a factor of 2.5, reduces artifacts, and increases the PSNR values by 13 dB.

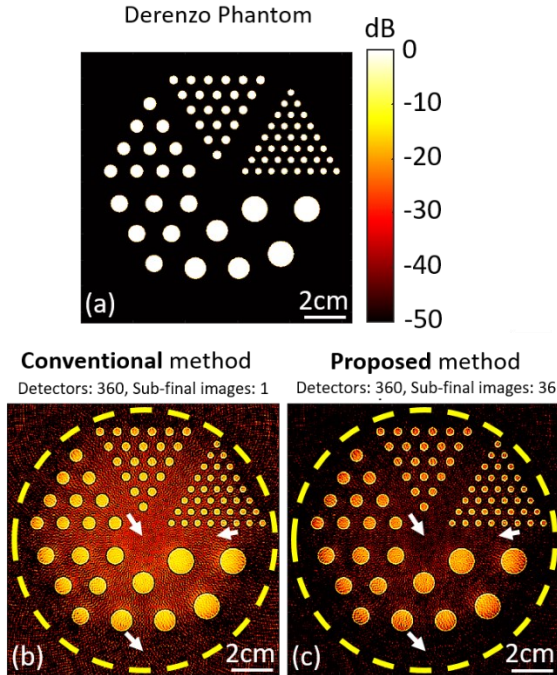


Fig. 2. The numerical results of Derenzo phantom. The yellow dashed line indicate the location of the transducer array. (a) The schematic of the Derenzo phantom. (b, c) The reconstructed image by the conventional and proposed method, respectively.

2) Case 2: Arteries phantom

Fig. 3(a) shows the Arteries phantom, and the yellow dashed circles indicate the positions of the ATs. Fig. 3(b, c, and d) show the reconstructed images obtained with the conventional and proposed method. We performed the proposed method with two different numbers of sub-images;

1) two sub-images and 2) 36 sub-images. Even though the proposed method with two sub-images enhances the image's quality more than the conventional method, adding more sub-images yields even further enhancements [compare the white arrows in Fig. 3(b-d)]. Compared to the two sub-image scenarios, the proposed method with 36 sub-images suppresses more artifacts. Quantitatively speaking, the proposed approach with 36 sub-images achieves an SSIM value of 0.72. Comparing this to the proposed method with two sub-images and the traditional method yields improvements of two and four times, respectively. Furthermore, compared to the conventional method, the proposed method improved PSNR by 11 dB (see TABLE I for other quantitative results).

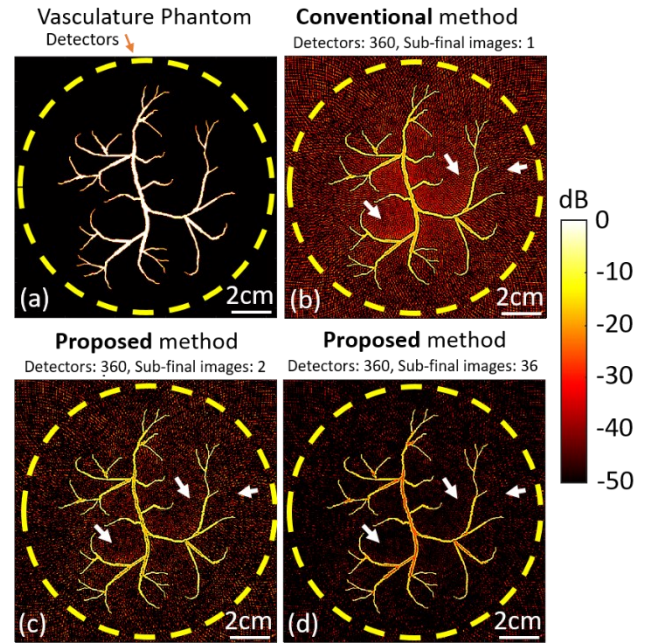


Fig. 3. The numerical results of Arteries phantom. The yellow dashed line indicate the location of the transducer array. (a) The schematic of the Arteries phantom. (b, c, d) The reconstructed image by the conventional and proposed method with 2 and 36 sub-images, respectively.

TABLE I
THE MEASURED QUANTITATIVE VALUES

Phantom Scenario	Derenzo		Arteries		
	Conv.	Proposed (NSI: 36)	Conv.	Proposed (NSI: 2)	Proposed (NSI: 36)
SSIM	0.29	0.76	0.20	0.41	0.78
PSNR[dB]	29	41	32	34	43

IV. DISCUSSION AND CONCLUSION

In this paper, we presented a coherent compounding method to improve image quality and suppress unwanted artifacts of the reconstructed image. We divided the total ATs into different groups, each of which containing a few ATs. Then, utilizing different sets of ATs for reconstructing each sub-image, results in obtaining several sub-images. The results show that due to selecting independent ATs to reconstruct each sub-image, streak artifacts appear in various positions in each sub-image. Therefore, the parts in the reconstructed image referring to structural information are maintained, and those referring to artifacts are suppressed as

we multiply a coherent compounding map of each sub-image pixel with signed values. The findings indicate that compared to the conventional method, the proposed method enhances the SNR value by 13 dB and the SSIM value up to 2.5-fold. Based on the results, the proposed method offers structural information with excellent contrast, making it a promising tool for breast/brain imaging.

REFERENCES

- [1] A. B. E. Attia *et al.*, "A review of clinical photoacoustic imaging: Current and future trends," *Photoacoustics*, vol. 16, p. 100144, Dec 2019, doi: 10.1016/j.pacs.2019.100144.
- [2] S. Manohar and M. Dantuma, "Current and future trends in photoacoustic breast imaging," *Photoacoustics*, vol. 16, p. 100134, Dec 2019, doi: 10.1016/j.pacs.2019.04.004.
- [3] S. Hakakzadeh *et al.*, "Finite Transducer Size Compensation in Two-Dimensional Photoacoustic Computed Tomography," in *2021 IEEE International Ultrasonics Symposium (IUS)*, 11-16 Sept. 2021 2021, pp. 1-3, doi: 10.1109/IUS52206.2021.9593757.
- [4] S. Hakakzadeh and Z. Kavehvas, "Blind Angle and Angular Range Detection in Planar and Limited-View Geometries for Photoacoustic Tomography," presented at the 2021 29th Iranian Conference on Electrical Engineering (ICEE), 2021.
- [5] P. J. van den Berg, K. Daoudi, and W. Steenbergen, "Pulsed photoacoustic flow imaging with a handheld system," *J Biomed Opt*, vol. 21, no. 2, p. 26004, Feb 2016, doi: 10.1117/1.JBO.21.2.026004.
- [6] S. Hakakzadeh and Z. Kavehvas, "Image Quality Equations for Focused Transducer in Circular Photoacoustic Computed Tomography," presented at the 2021 29th Iranian Conference on Electrical Engineering (ICEE), 2021.
- [7] M. Ai *et al.*, "Investigation of photoacoustic tomography reconstruction with a limited view from linear array," *J Biomed Opt*, vol. 26, no. 9, Sep 2021, doi: 10.1117/1.JBO.26.9.096009.
- [8] S. Hakakzadeh *et al.*, "Multi-angle data acquisition to compensate transducer finite size in photoacoustic tomography," *Photoacoustics*, vol. 27, p. 100373, Sep 2022, doi: 10.1016/j.pacs.2022.100373.
- [9] M. Xu and L. V. Wang, "Analytic explanation of spatial resolution related to bandwidth and detector aperture size in thermoacoustic or photoacoustic reconstruction," *Physical Review E*, vol. 67, no. 5, 2003, doi: 10.1103/physreve.67.056605.
- [10] S. Hakakzadeh, M. Amjadian, Y. Zhang, S. M. Mostafavi, Z. Kavehvas, and L. Wang, "Signal restoration algorithm for photoacoustic imaging systems," *Biomed Opt Express*, vol. 14, no. 2, pp. 651-666, Feb 1 2023, doi: 10.1364/BOE.480842.
- [11] S. Hakakzadeh, S. M. Mostafavi, and Z. Kavehvas, "Unipolar Back-Projection Algorithm for Photoacoustic Tomography," presented at the 2022 IEEE International Ultrasonics Symposium (IUS), 2022.
- [12] L. Li, L. Zhu, Y. Shen, and L. V. Wang, "Multiview Hilbert transformation in full-ring transducer array-based photoacoustic computed tomography," *J Biomed Opt*, vol. 22, no. 7, p. 76017, Jul 1 2017, doi: 10.1117/1.JBO.22.7.076017.
- [13] S. Hakakzadeh, P. Rajendran, V. A. Nili, Z. Kavehvas, and M. Pramanik, "A Spatial-Domain Factor for Sparse-Sampling Circular-View Photoacoustic Tomography," *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 29, no. 4: Biophotonics, pp. 1-9, 2023, doi: 10.1109/jstqe.2022.3229622.
- [14] L. Li *et al.*, "Single-impulse panoramic photoacoustic computed tomography of small-animal whole-body dynamics at high spatiotemporal resolution," *Nature Biomedical Engineering*, vol. 1, no. 5, p. 0071, 2017/05/10 2017, doi: 10.1038/s41551-017-0071.
- [15] P. Hu, L. Li, L. Lin, and L. V. Wang, "Spatiotemporal Antialiasing in Photoacoustic Computed Tomography," *IEEE Trans Med Imaging*, vol. 39, no. 11, pp. 3535-3547, Nov 2020, doi: 10.1109/TMI.2020.2998509.
- [16] C. Cai, X. Wang, K. Si, J. Qian, J. Luo, and C. Ma, "Streak artifact suppression in photoacoustic computed tomography using adaptive back projection," *Biomed Opt Express*, vol. 10, no. 9, pp. 4803-4814, Sep 1 2019, doi: 10.1364/BOE.10.004803.
- [17] S. Hakakzadeh, Z. Kavehvas, and M. Pramanik, "Artifact Removal Factor for Circular-view Photoacoustic Tomography," presented at the 2022 IEEE International Ultrasonics Symposium (IUS), 2022.
- [18] S. Hakakzadeh, S. M. Mostafavi, M. Amjadian, and Z. Kavehvas, "Adapted Coherent Weighting in Photoacoustic Tomography," presented at the 2021 28th National and 6th International Iranian Conference on Biomedical Engineering (ICBME), 2021.
- [19] S. Hakakzadeh and Z. Kavehvas, "A Hilbert-based Coherence Factor for Photoacoustic Imaging," presented at the 2022 29th National and 7th International Iranian Conference on Biomedical Engineering (ICBME), 2022.
- [20] M. A. Araque Caballero, J. Gateau, X.-L. Dean-Ben, and V. Ntziachristos, "Model-Based Optoacoustic Image Reconstruction of Large Three-Dimensional Tomographic Datasets Acquired With an Array of Directional Detectors," *IEEE Transactions on Medical Imaging*, vol. 33, no. 2, pp. 433-443, 2014, doi: 10.1109/tmi.2013.2286546.
- [21] L. Ding, D. Razansky, and X. L. Dean-Ben, "Model-Based Reconstruction of Large Three-Dimensional Optoacoustic Datasets," *IEEE Trans Med Imaging*, vol. 39, no. 9, pp. 2931-2940, Sep 2020, doi: 10.1109/TMI.2020.2981835.
- [22] C. Lutzweiler, X. L. Deán-Ben, and D. Razansky, "Expediting model-based optoacoustic reconstructions with tomographic symmetries," *Medical Physics*, vol. 41, no. 1, p. 013302, 2014, doi: <https://doi.org/10.1118/1.4846055>.
- [23] A. DiSpirito, 3rd, T. Vu, M. Pramanik, and J. Yao, "Sounding out the hidden data: A concise review of deep learning in photoacoustic imaging," *Exp Biol Med (Maywood)*, vol. 246, no. 12, pp. 1355-1367, Jun 2021, doi: 10.1177/15353702211000310.
- [24] N. Awasthi, G. Jain, S. K. Kalva, M. Pramanik, and P. K. Yalavarthy, "Deep Neural Network-Based Sinogram Super-Resolution and Bandwidth Enhancement for Limited-Data Photoacoustic Tomography," *IEEE Trans Ultrason Ferroelectr Freq Control*, vol. 67, no. 12, pp. 2660-2673, Dec 2020, doi: 10.1109/TUFFC.2020.2977210.
- [25] M. Lu *et al.*, "Artifact removal in photoacoustic tomography with an unsupervised method," *Biomed Opt Express*, vol. 12, no. 10, pp. 6284-6299, Oct 1 2021, doi: 10.1364/BOE.434172.
- [26] N. Davoudi, X. L. Deán-Ben, and D. Razansky, "Deep learning optoacoustic tomography with sparse data," *Nature Machine Intelligence*, vol. 1, no. 10, pp. 453-460, 2019, doi: 10.1038/s42256-019-0095-3.
- [27] N. Davoudi, B. Lafci, A. Ozbek, X. L. Dean-Ben, and D. Razansky, "Deep learning of image- and time-domain data enhances the visibility of structures in optoacoustic tomography," *Opt Lett*, vol. 46, no. 13, pp. 3029-3032, Jul 1 2021, doi: 10.1364/OL.424571.
- [28] S. Hakakzadeh, S. M. Mostafavi, and Z. Kavehvas, "Fast Adapted Delay and Sum Reconstruction Algorithm in Circular Photoacoustic Tomography," presented at the 2022 30th International Conference on Electrical Engineering (ICEE), 2022.
- [29] B. Treeby and B. Cox, "k-Wave: MATLAB toolbox for the simulation and reconstruction of photoacoustic wave fields," *Journal of Biomedical Optics*, vol. 15, no. 2, p. 021314, 2010. [Online]. Available: <https://doi.org/10.1117/1.3360308>.